# Optimizing energy conversion in nanoelectronic devices exploiting nonthermal resources

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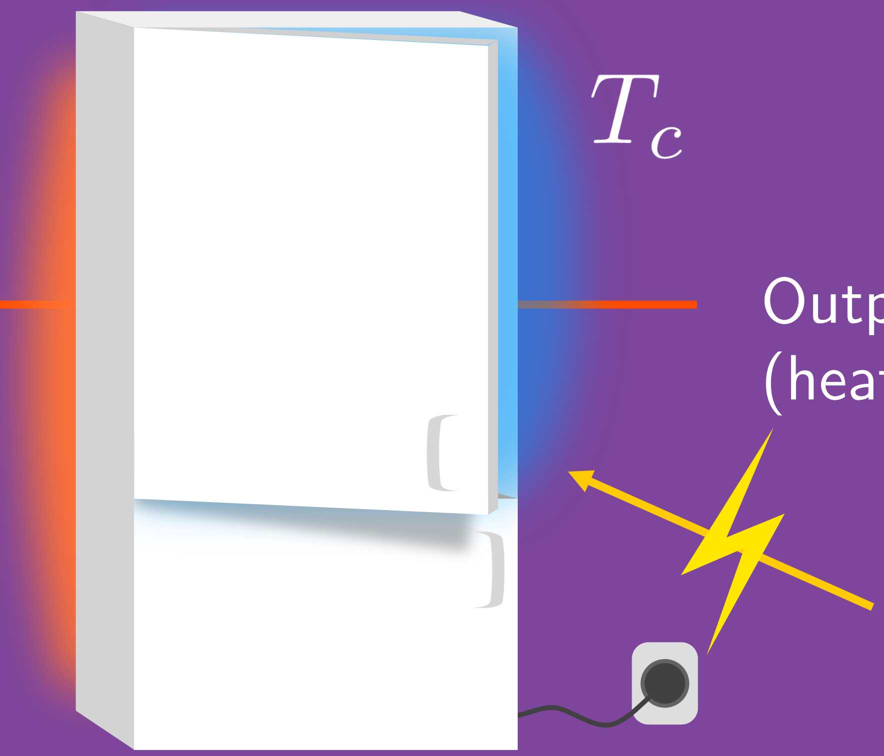
"Optimizing energy conversion with nonthermal resources for steady-state quantum devices", to be submitted, Elsa Danielsson, Henning Kirchberg, Janine Splettstoesser

# The refrigerator

$$\eta = \frac{\text{Cooling power}}{\text{Electric power}}$$

$T_h$

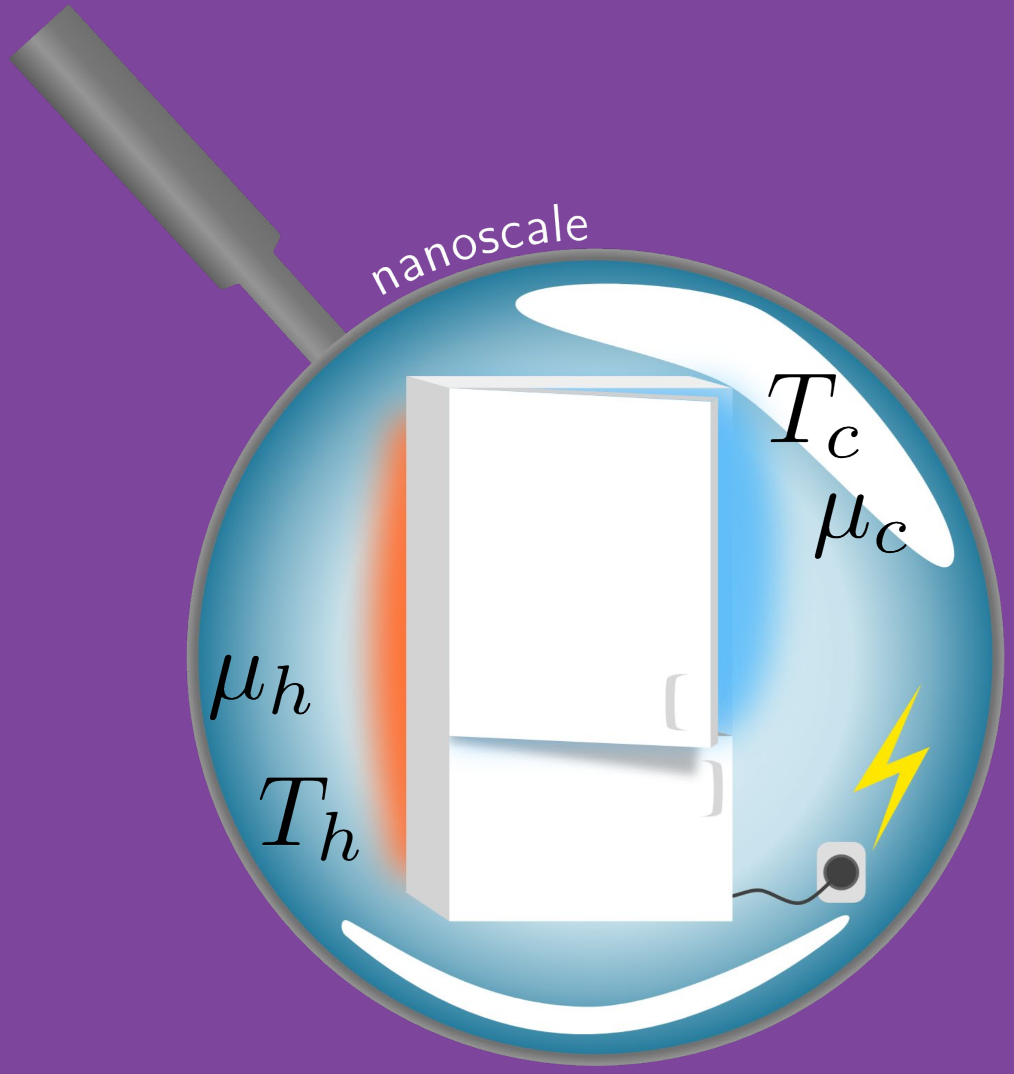


$T_c$

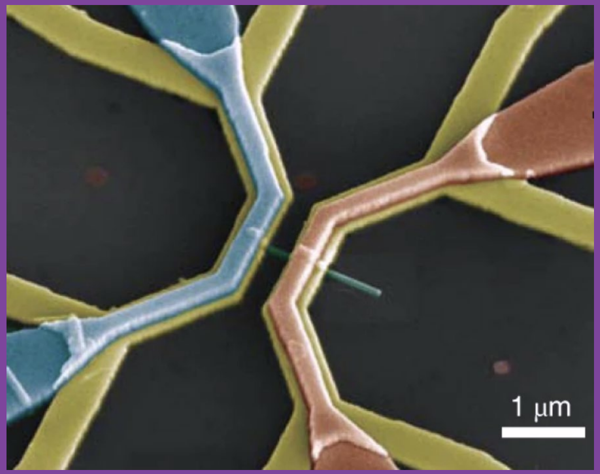
Output = Cooling power (heat transfer)

Resource = Electric power

# The refrigerator – but smaller




Quantum dot heat engine



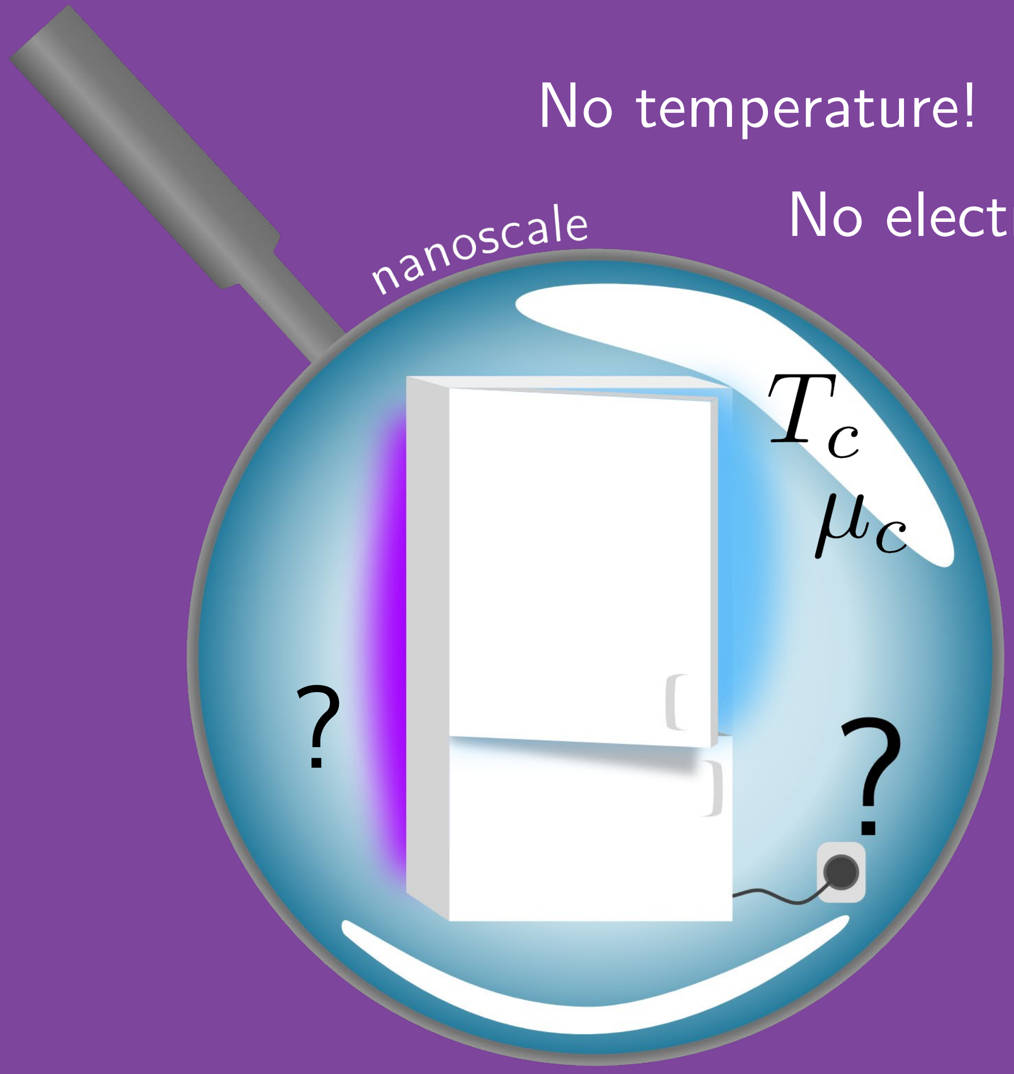

M. Josefsson et al.: Nat. Nanotechnol. 13, 920 (2018)

- Thermoelectric effect
- Steady-state

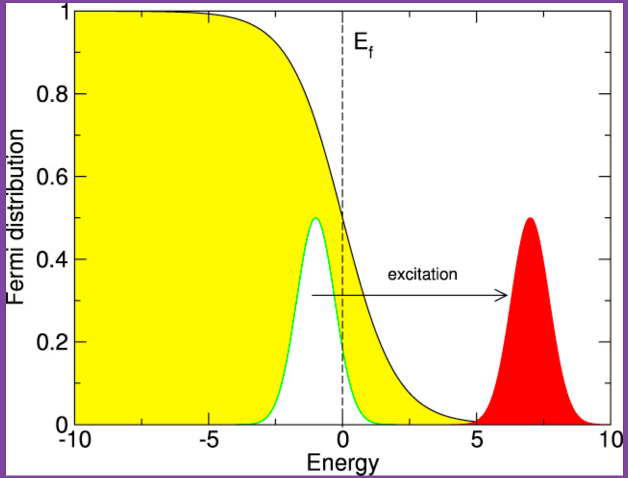
# The refrigerator – but nonthermal

No temperature!

No electrochemical potential!

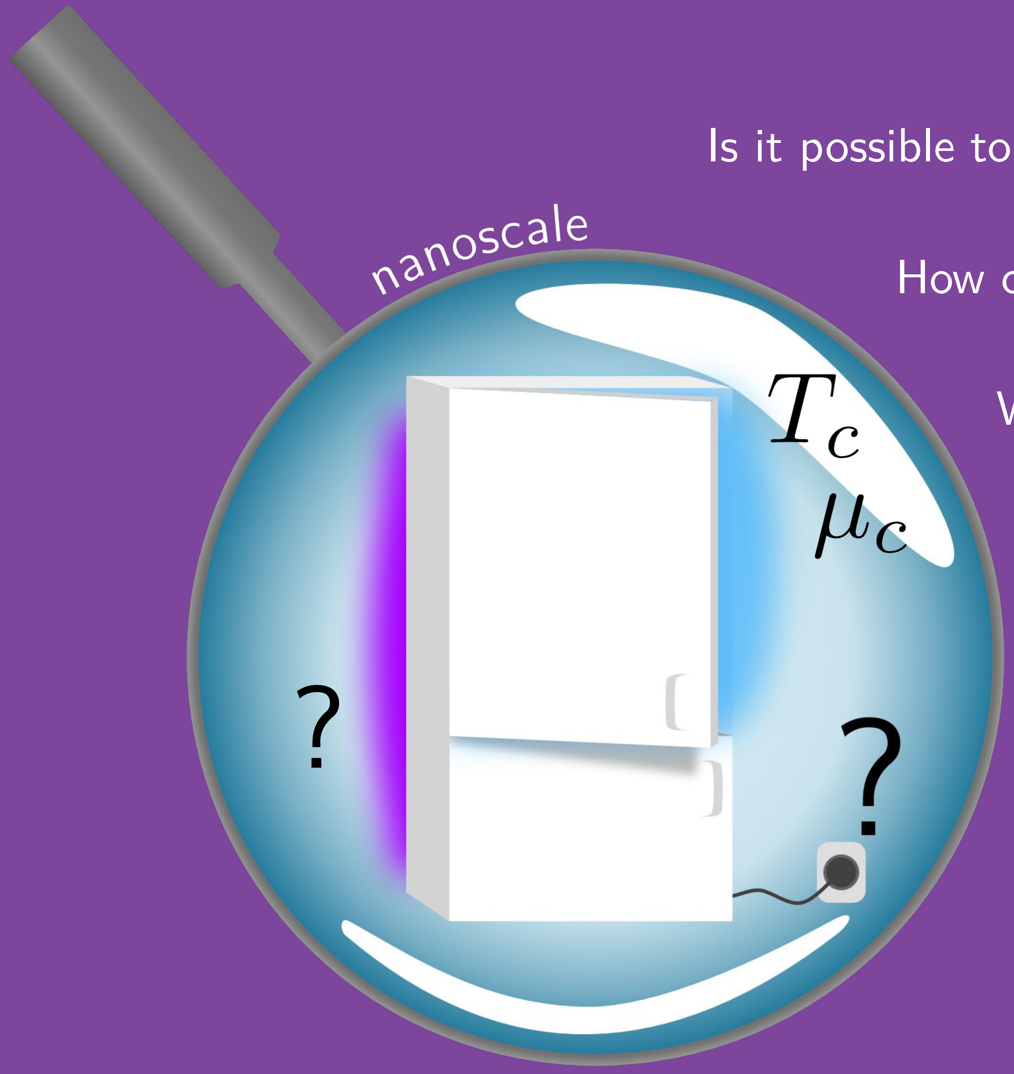



Nonthermal distribution
from irradiation




G. P. Zhang et al.: J. Phys.:
Condens. Matter 25, 366002 (2013)

# The refrigerator – but nonthermal




Is it possible to cool with a nonthermal reservoir as a resource?

How can we define the performance of the refrigerator?

What is the resource expended?

How do we make the **best-performing refrigerator?**

We will consider the **entropy production** in the nonthermal bath as the **resource!**

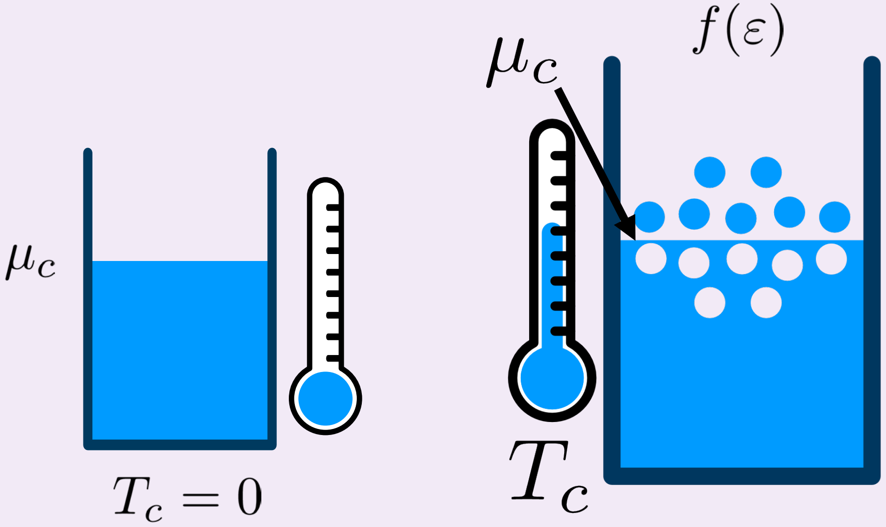
- Proportional to heat current
- Second Law

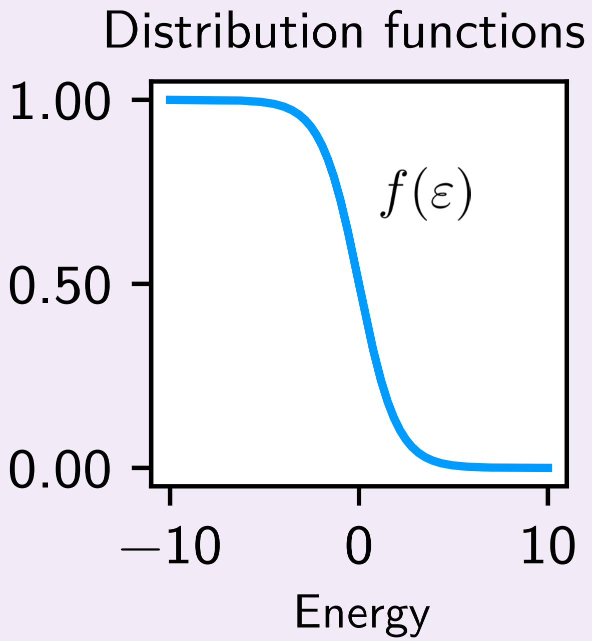
# Outline

- Introduction
- Thermoelectric effect with two terminals
- Optimization methods for general currents
- Example for cooling with entropy
- Conclusion and questions

# Thermoelectric effect with two terminals

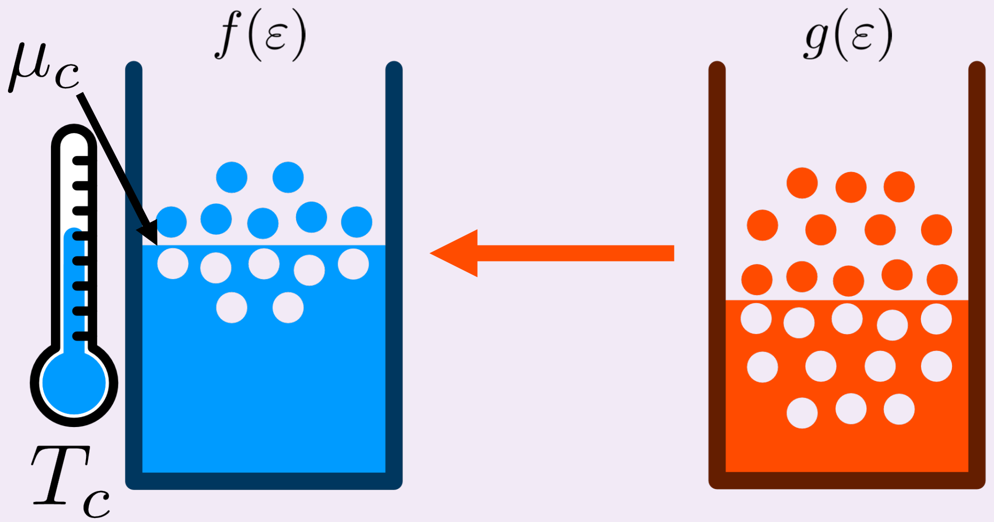
# Electron reservoir



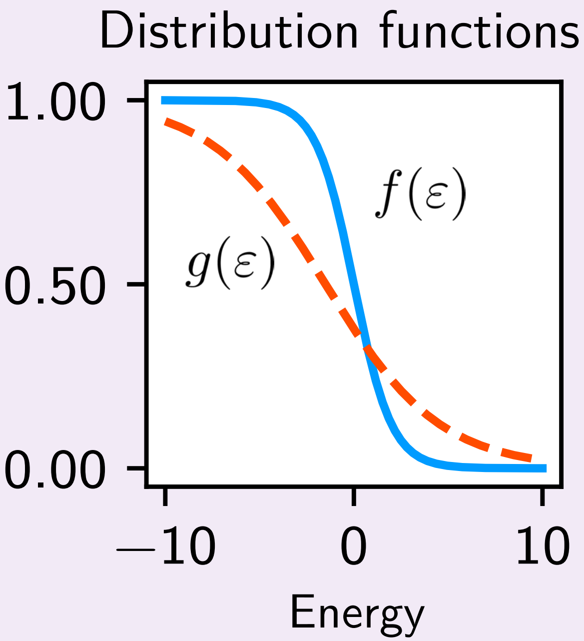




Review: G. Benenti et al.: Phys. Rep. 694, 1 (2017)
R. S. Whitney, Phys. Rev. B, 91 (2015)
A. M. Timpanaro et al.: Phys. Rev. B 111, 014301 (2025)

# Introducing another bath




Current flows from more
to less occupied





Review: G. Benenti et al.: Phys. Rep. 694, 1 (2017)
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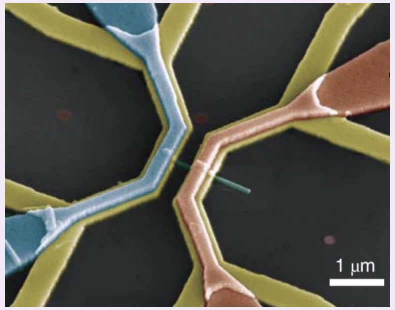
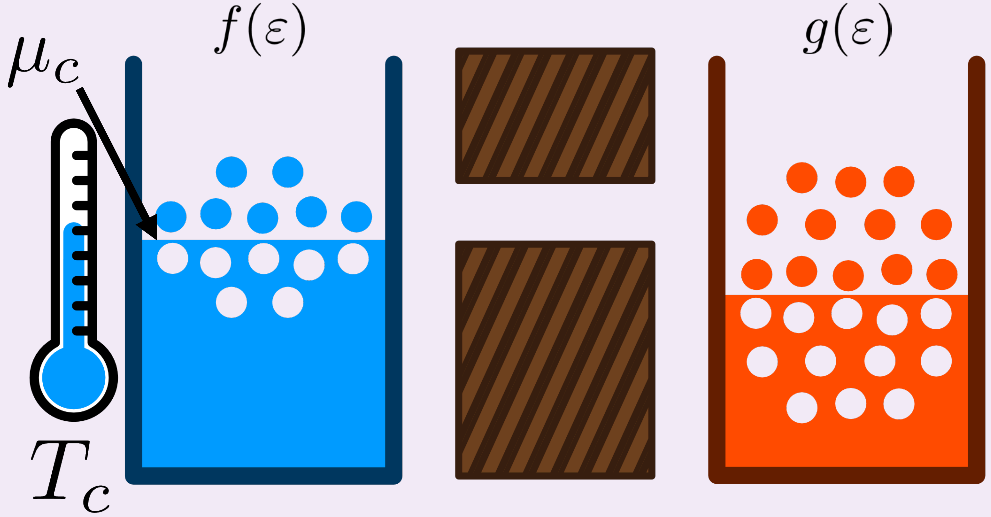
A. M. Timpanaro et al.: Phys. Rev. B 111, 014301 (2025)

# Structured electron transfer

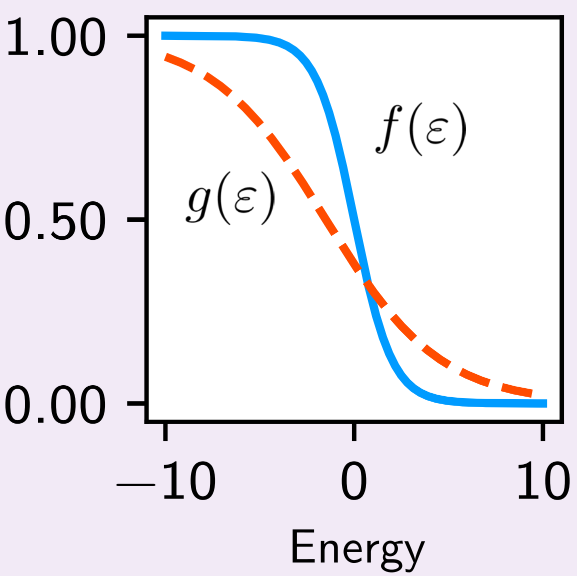



M. Josefsson et al.: Nat. Nanotechnol. 13, 920 (2018)




Nanostructure coupling the two baths

Distribution functions

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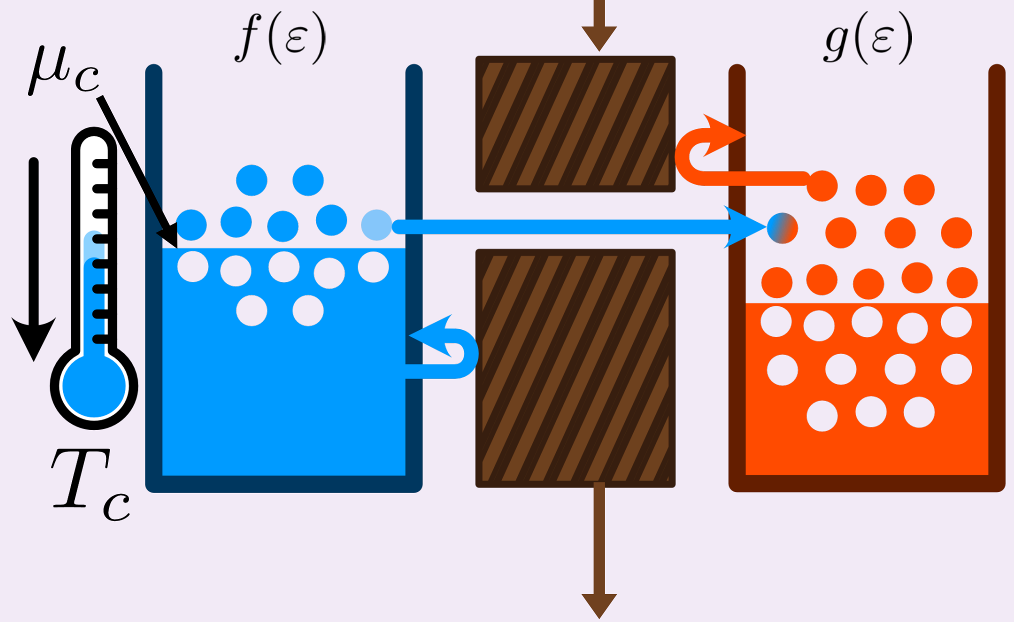
R. S. Whitney, Phys. Rev. B, 91 (2015)
A. M. Timpanaro et al.: Phys. Rev. B 111, 014301 (2025)

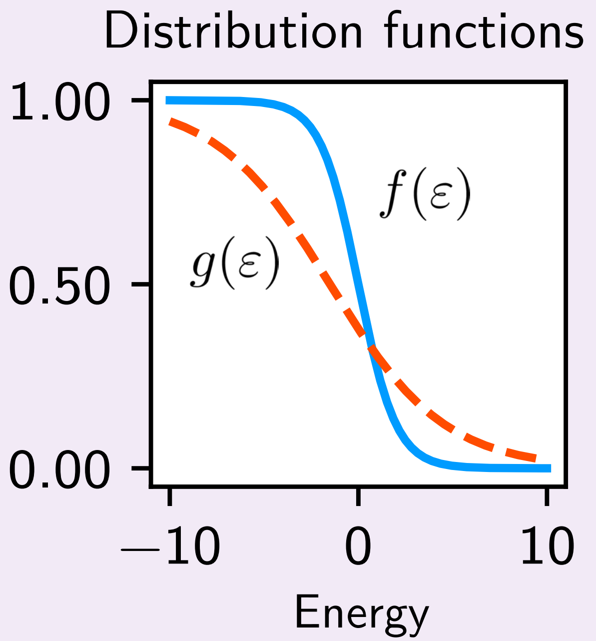
# Steady-state thermoelectric

$$I^x = \int d\varepsilon\, x(\varepsilon) \boxed{\mathcal{D}(\varepsilon)} (f(\varepsilon) - g(\varepsilon))$$

no electron-electron interaction

$x(\varepsilon) = \varepsilon - \mu_c$



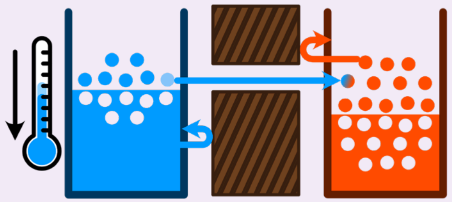




Transmission function defined between 0 and 1 – between full reflection and full transmission

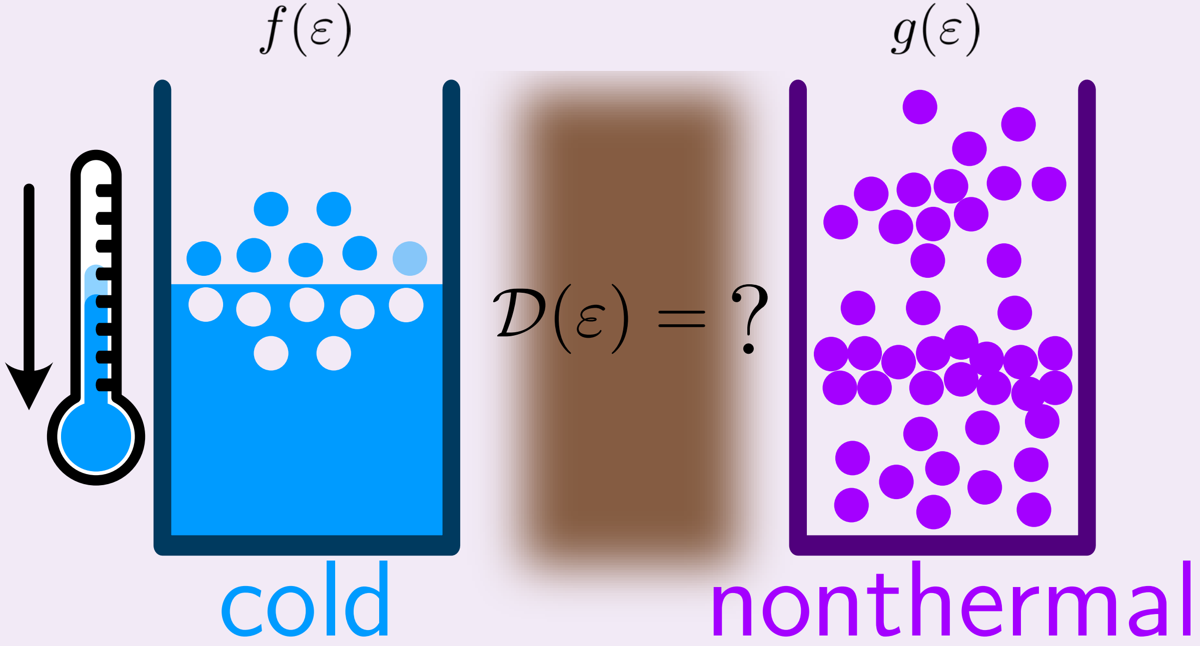
Review: G. Benenti et al.: Phys. Rep. 694, 1 (2017)
R. S. Whitney, Phys. Rev. B, 91 (2015)
A. M. Timpanaro et al.: Phys. Rev. B 111, 014301 (2025)

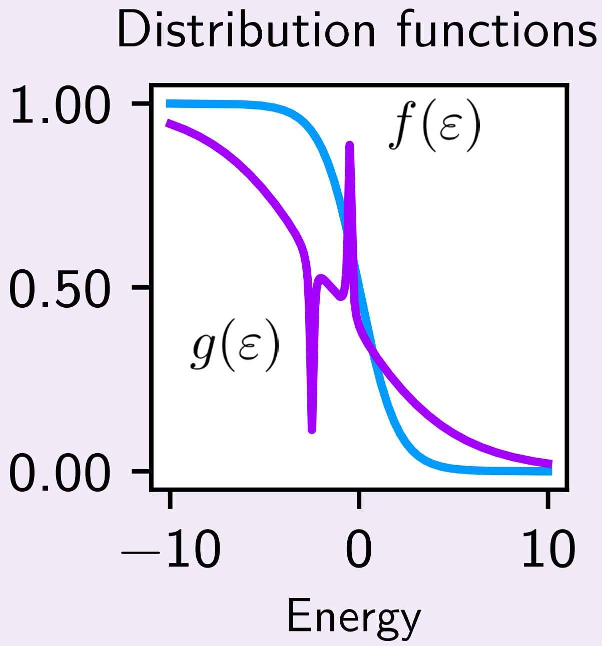
# Nonthermal cooling

Replace hot thermal bath with **nonthermal** bath. What changes?



$$I^x = \int d\varepsilon\, x(\varepsilon)\mathcal{D}(\varepsilon)(f(\varepsilon) - g(\varepsilon))$$



# Optimization methods for general currents

Presented: Maximize output and maximize efficiency

Additional in paper: Maximize precision and maximize trade-off relation (TUR)

# Maximized output

**Goal**: Cool as much as possible!

**Solution**: Transmit all electrons that contribute to cooling and block the rest

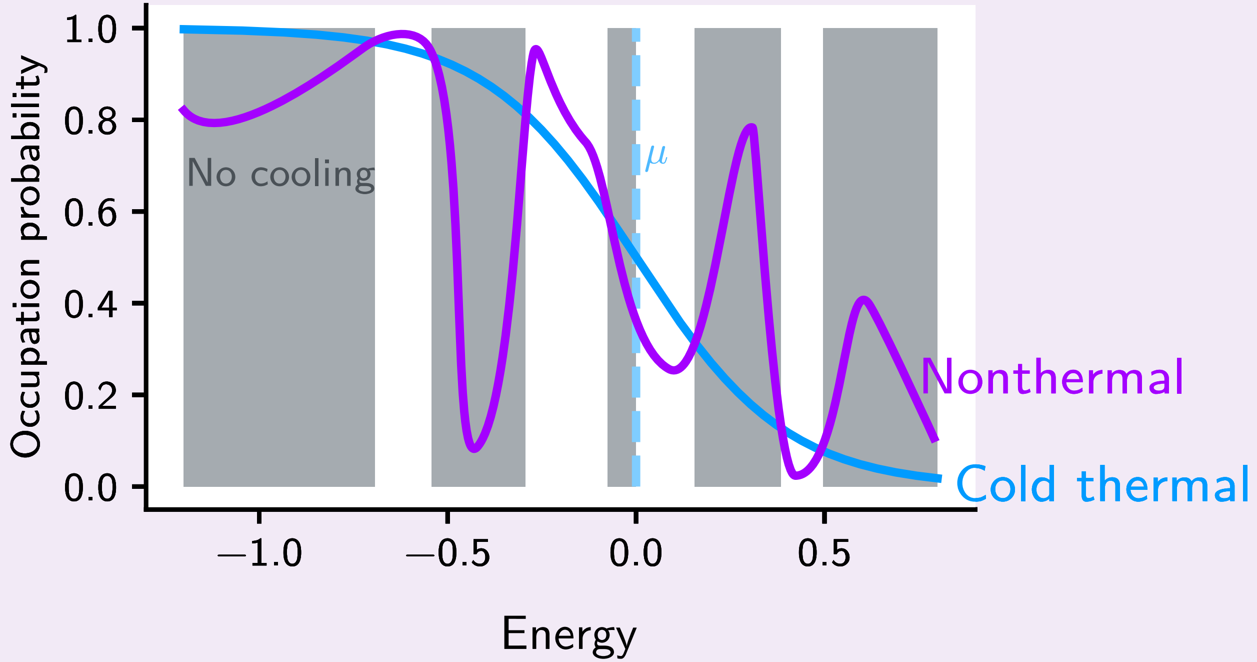
$$I^x = \int d\varepsilon\, x(\varepsilon)\mathcal{D}(\varepsilon)(f(\varepsilon) - g(\varepsilon))$$

Spectral current at full transmission

$$i^x(\varepsilon) = x(\varepsilon)(f(\varepsilon) - g(\varepsilon))$$

**Positive = cooling**

Transmission for maximized cooling with arbitrary nonthermal resource



# Maximized output

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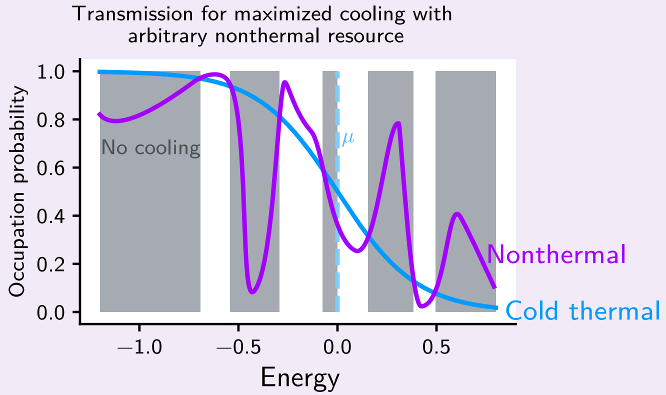
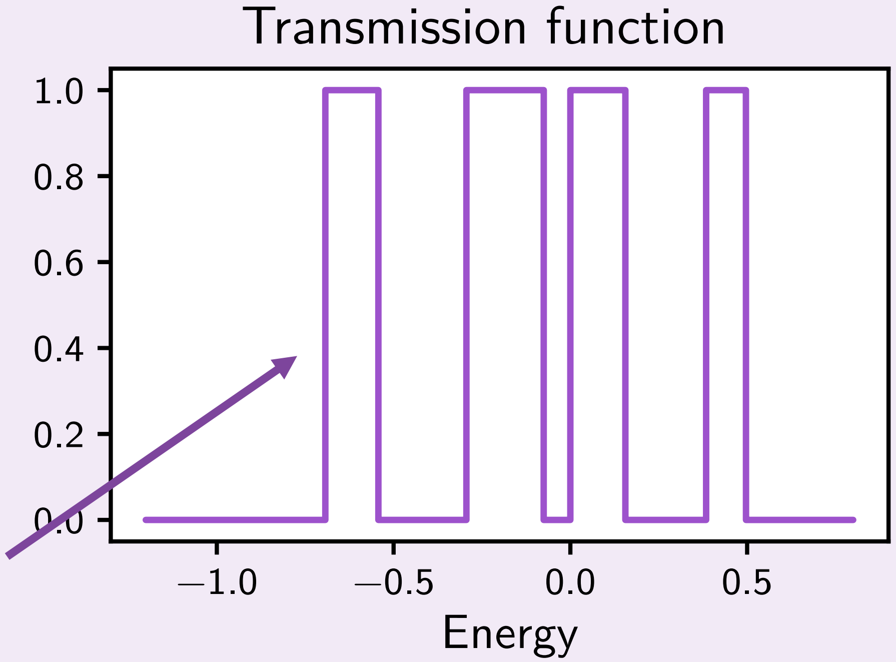



Series of boxcars

# Optimize for efficiency

**Goal**: Get the highest possible efficiency for a given cooling power

Efficiency

$$\eta = \frac{I^x}{I^y}$$

Output current (Cooling power/entropy reduction in cold bath)

Resource current (entropy production in nonthermal bath)

**Solution**: Transmit the electrons that contribute *most efficienctly* to the cooling power and block the rest

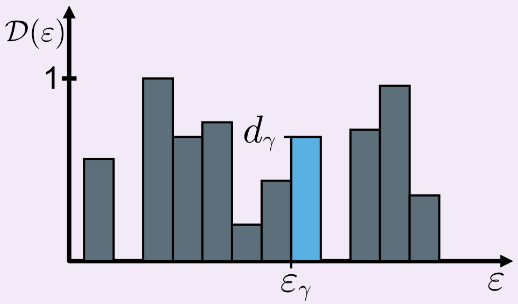
# Linear transmission dependency

$$\eta = \frac{I^x}{I^y}$$

Discrete version of current

$$I^x = \sum_\gamma I^x_\gamma(d_\gamma)$$

Sliced transmission function



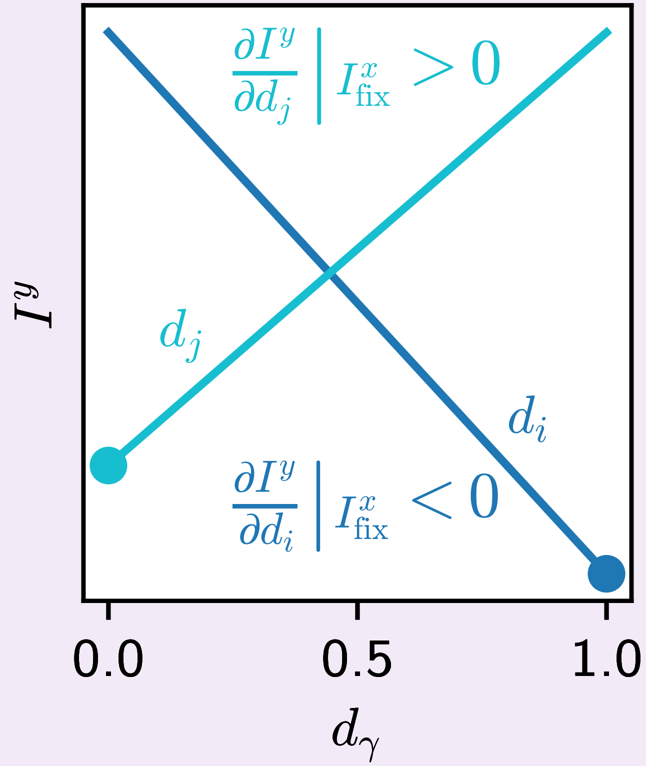

Constraint equation

$$I^y = \sum_\gamma I^y_\gamma(d_\gamma) + \lambda\big(I^x_{\text{fix}} - \sum_\gamma I^x_\gamma(d_\gamma)\big)$$

Minimize

What happens when changing the transmission in one slice?

$$\left.\frac{\partial I^y}{\partial d_\gamma}\right|_{I^x_{\text{fix}}} = \delta\varepsilon(y_\gamma - \lambda x_\gamma)(f_\gamma - g_\gamma)$$



# Linear transmission dependency

$$\boxed{\eta = \frac{I^x}{I^y}}$$

Condition for full transmission

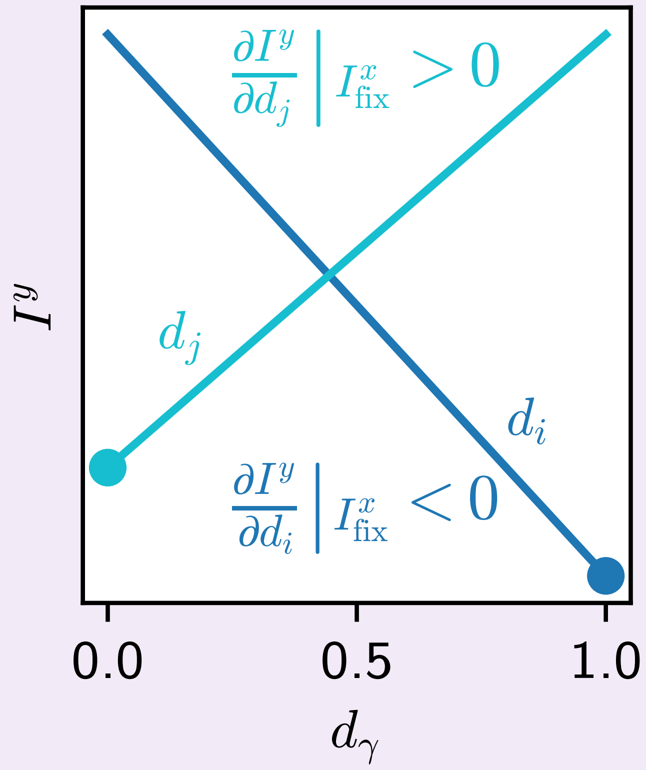
$$\left.\frac{\partial I^y}{\partial d_\gamma}\right|_{I^x_{\mathrm{fix}}} = \delta\varepsilon(y_\gamma - \lambda x_\gamma)(f_\gamma - g_\gamma) < 0$$

Continous limit, functional variation...

$$\mathcal{D}^{\eta}_{\mathrm{opt}}(\varepsilon) = \Theta(-(y(\varepsilon) - \lambda x(\varepsilon))(f(\varepsilon) - g(\varepsilon)))$$

Stepfunction!

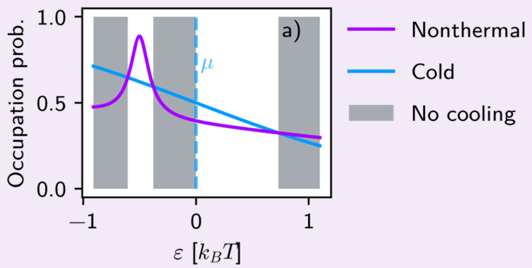
**The optimal transmission function is a series of boxcars**



# Characteristic efficiency

**Solution**: Transmit the electrons that contribute *most efficienctly* to the cooling power and block the rest

$$\mathcal{D}^{\eta}_{\text{opt}}(\varepsilon) = \Theta(-(y(\varepsilon) - \lambda x(\varepsilon))(f(\varepsilon) - g(\varepsilon)))$$

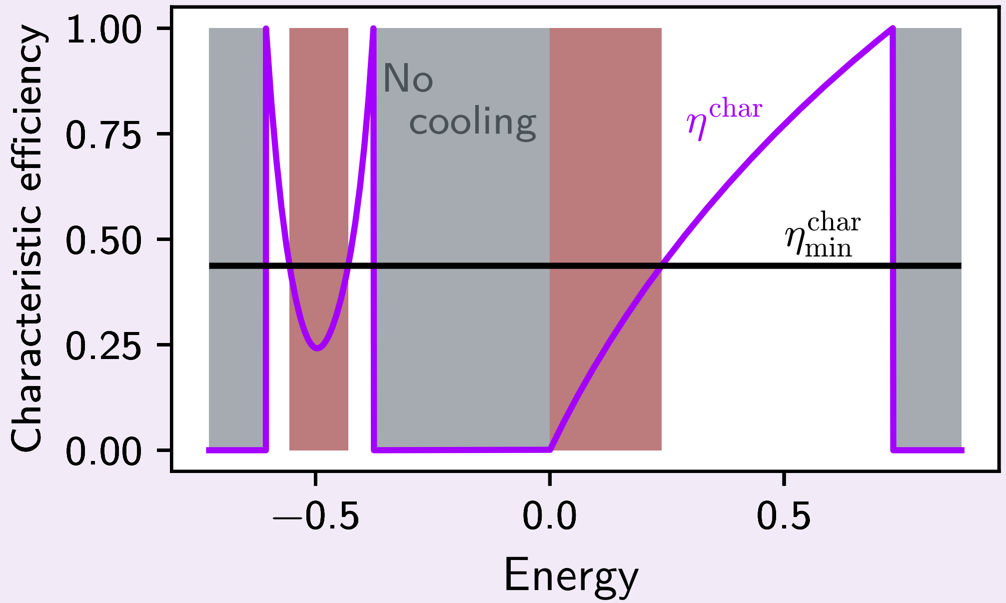



$$\eta^{\text{char}}(\varepsilon) \equiv \frac{i^x(\varepsilon)}{i^y(\varepsilon)}\Theta(i^x(\varepsilon))\Theta(i^y(\varepsilon))$$

For well-defined currents that do not violate the Second Law (both spectral currents must be positive):

$$\mathcal{D}^{\eta}_{\text{opt}}(\varepsilon) = \Theta(\eta^{\text{char}}(\varepsilon) - \eta^{\text{char}}_{\text{min}})$$

$$\eta^{\text{char}}_{\text{min}} \equiv 1/\lambda$$



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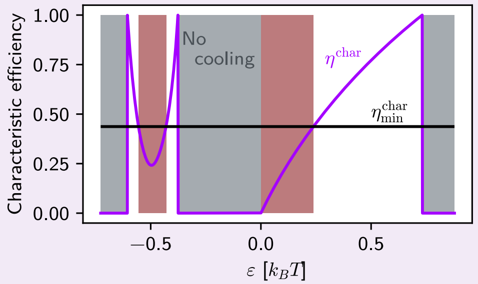
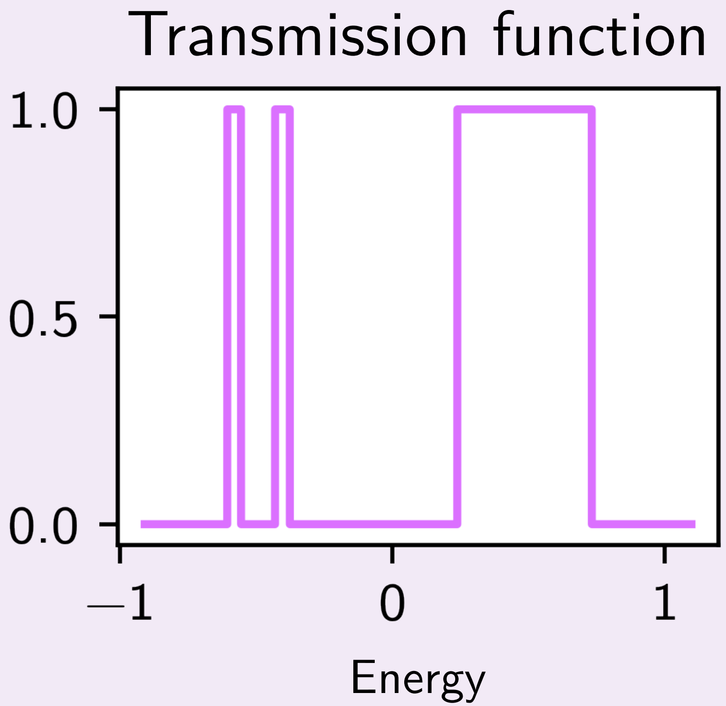


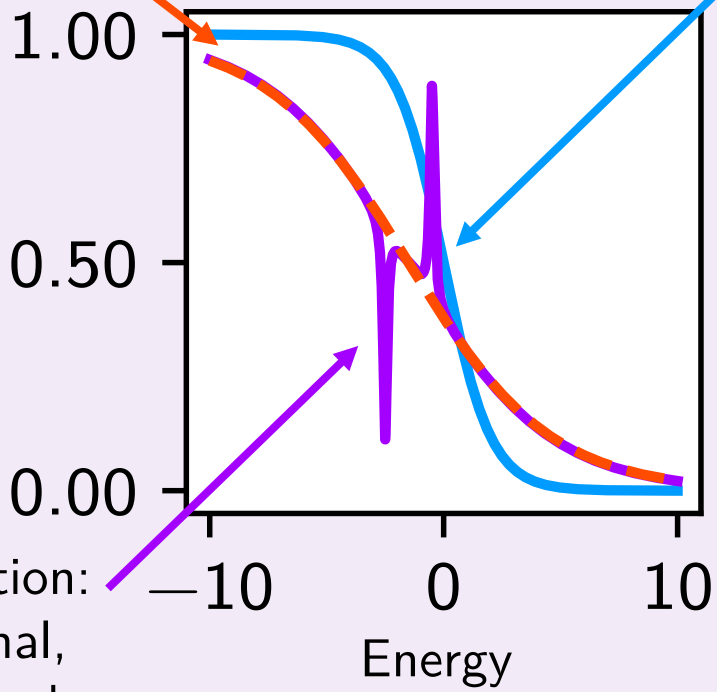



# Example for cooling with entropy

# The distributions

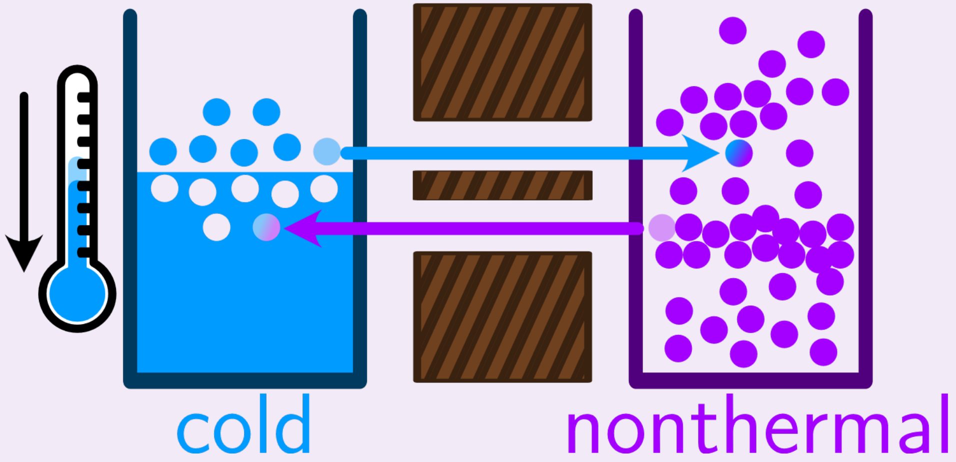
Büttiker probe to compare with

Electrochemical potential of cold bath is at 0




Nonthermal distribution: underlying hot thermal, add Lorentzian dip and peak

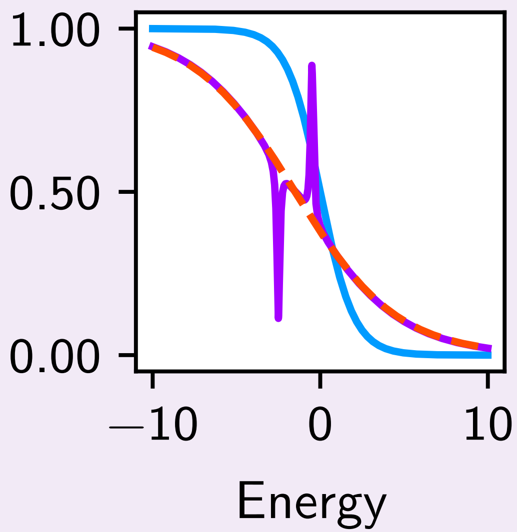
Nonthermal refrigerator

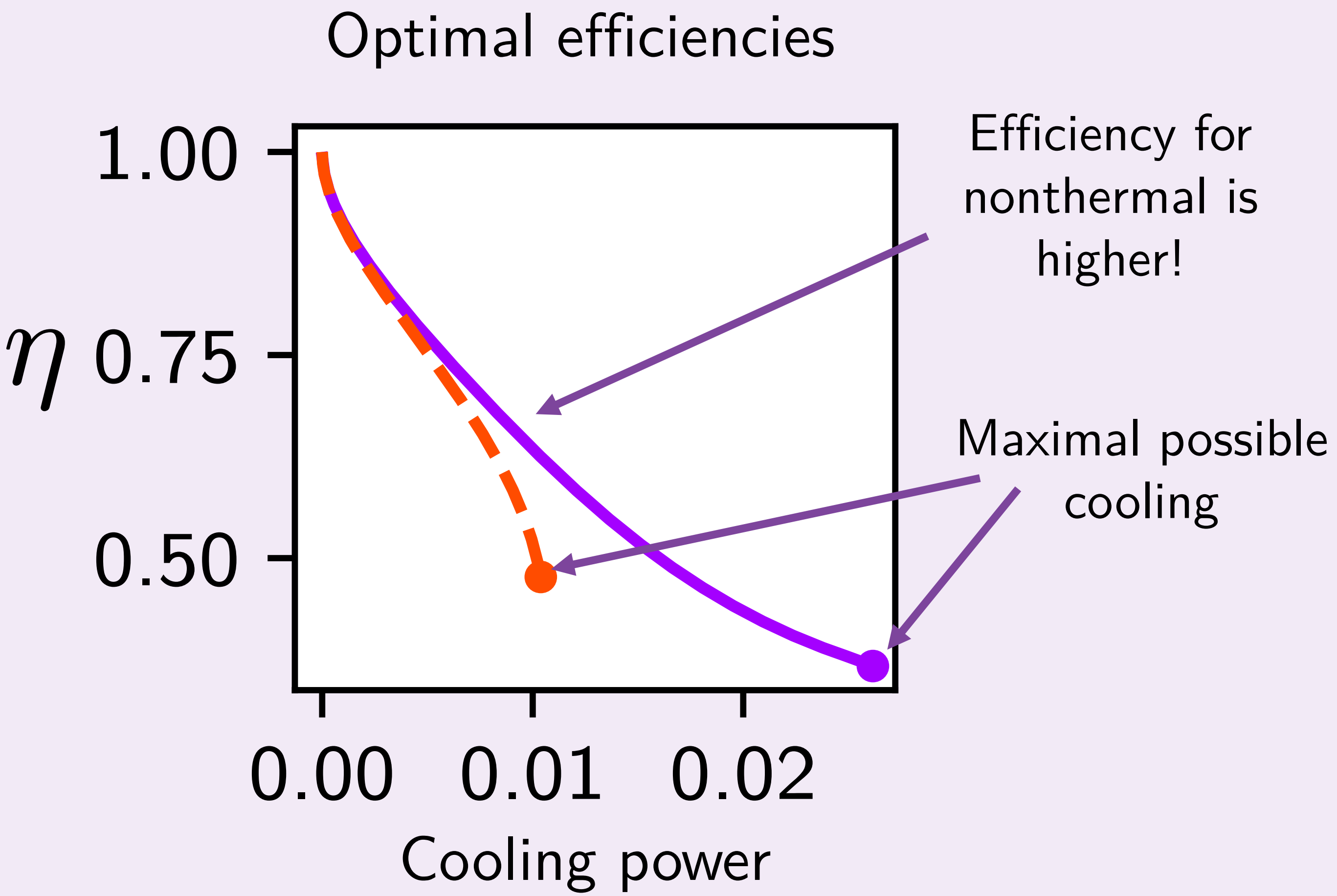



Büttiker probe: M. Büttiker: Phys. Rev. B 33, 3020 (1986); Phys. Rev. B 32, 1846 (1985)

# Comparing thermal and nonthermal

Find the optimal transmission function for efficiency, over the entire range of cooling power



# Conclusion

# Conclusion

- **Nonthermal** reservoirs can be exploited for energy conversion, and perhaps even **better than thermal** counterparts.
- This is shown by optimizing the transmission function for efficiency, which is **achievable for any task**, and it turns out it is **always a series of boxcars**.
- The solution follows the principle: Transmit the electrons that are **most beneficial** and **block the rest**
- All of this can be applied to **precision** and **trade-off** optimizations too!

"Optimizing energy conversion with nonthermal resources for steady-state quantum devices", to be submitted, Elsa Danielsson, Henning Kirchberg, Janine Splettstoesser

# Group: Dynamics and thermodynamics of nanoscale devices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# Optimize for precision

**Goal**: Maximize signal-to-noise ratio

$$\mathrm{SNR} = \frac{I^x}{S^x}$$

**Solution**: Transmit the electrons that contribute *most precisely* to the cooling power and block the rest

Quadratic dependence in noise $\mathcal{D}(1-\mathcal{D})$

Goes to 0 whenever transmission is 0 or 1
Optimization reduced to linear like before

# Optimize for precision

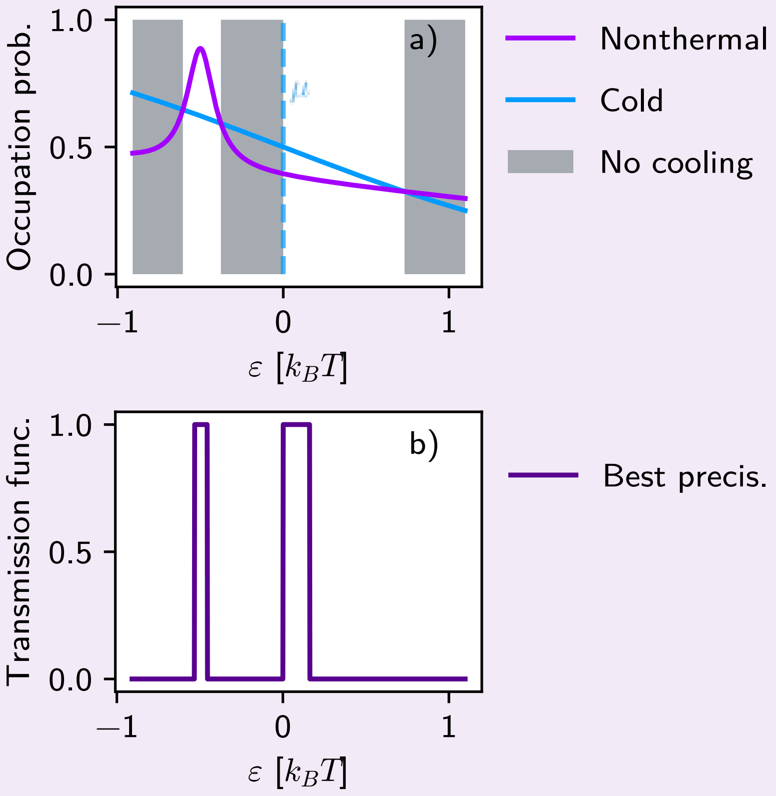
**Goal**: Maximize signal-to-noise ratio

$$\mathrm{SNR} = \frac{I^x}{S^x}$$

**Solution**: Transmit the electrons that contribute *most precisely* to the cooling power and block the rest

$$\mathrm{SNR}^{\mathrm{char}}(\varepsilon) = \frac{i^x(\varepsilon)}{s^x(\varepsilon)}\Theta(i^x(\varepsilon))$$

$$\mathcal{D}^S_{\mathrm{opt}}(\varepsilon) = \Theta(\mathrm{SNR}^{\mathrm{char}}(\varepsilon) - \mathrm{SNR}^{\mathrm{char}}_{\mathrm{min}})$$



# Optimize for trade-off

**Goal**: Minimize TUR-like trade-off relation

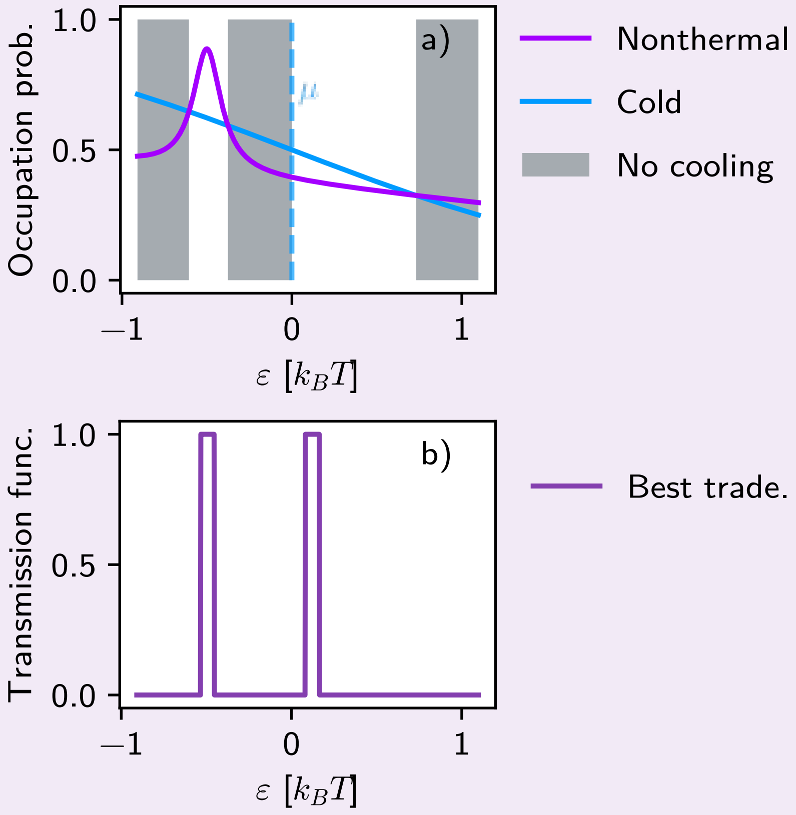
$$A = \frac{S^x I^y}{(I^x)^2}$$

**Solution**: Transmit the electrons that contribute *the least to A* and block the rest

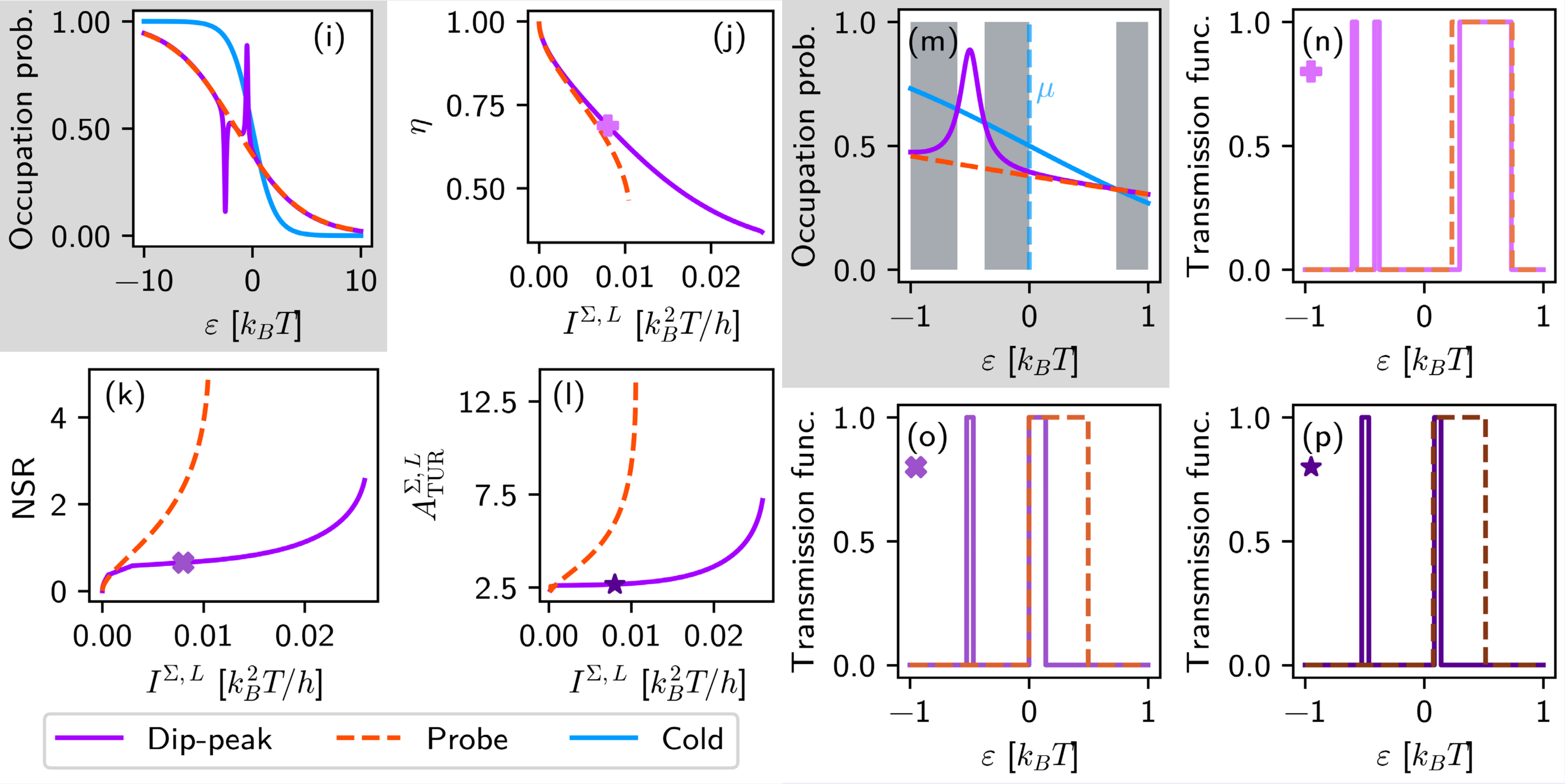
$$\delta(S^x I^y) = I^y \delta S^x + S^x \delta I^y$$

$$\mathcal{D}_{\text{opt}}^{\text{trade}}(\varepsilon) = \Theta(-I^y \ldots - S^x \ldots + \lambda \ldots)$$

Need to solve two self-consistency equations and one constraint!

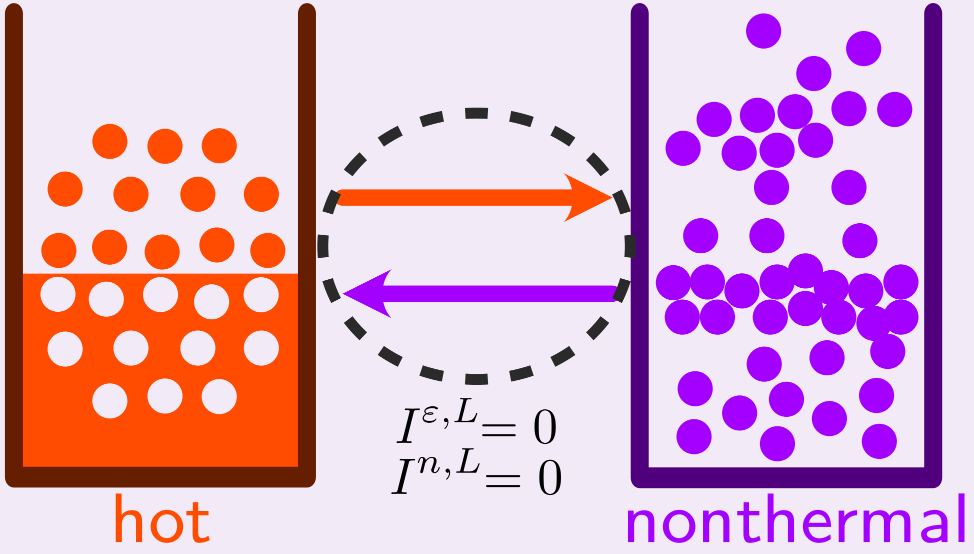


# Comparing thermal and nonthermal



# Thermal probe

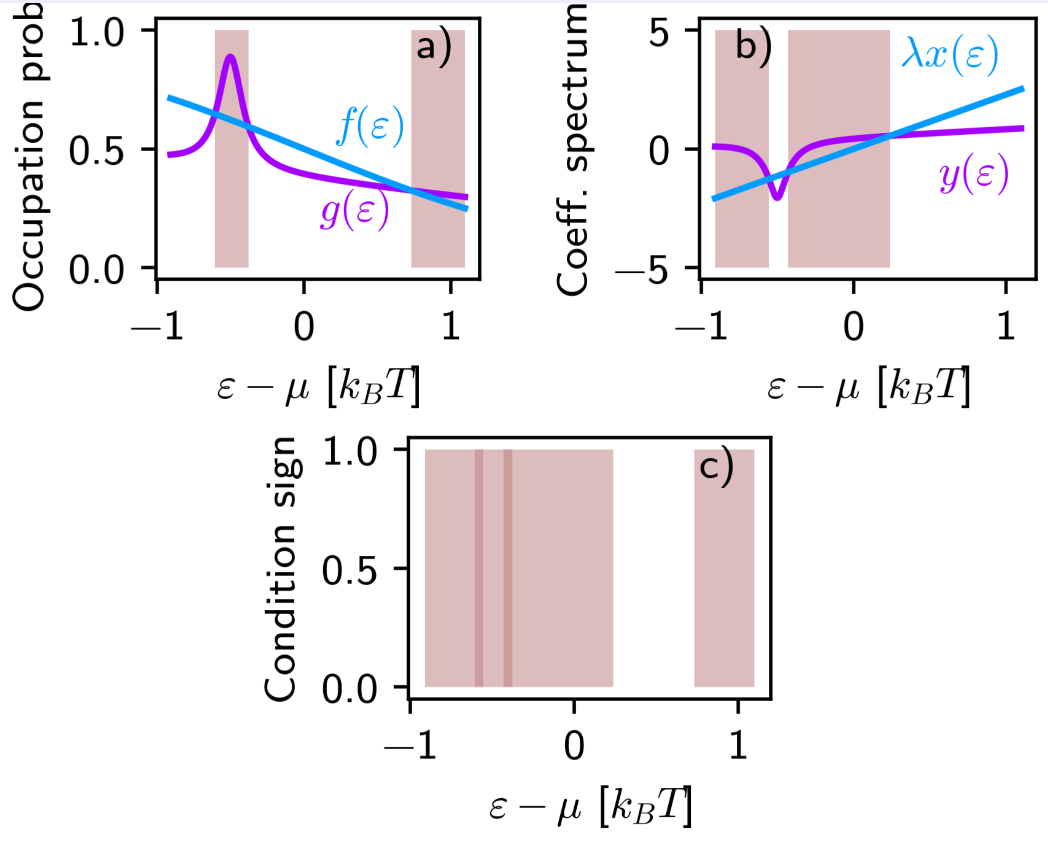
Need to compare the performance of the nonthermal reservoir with something

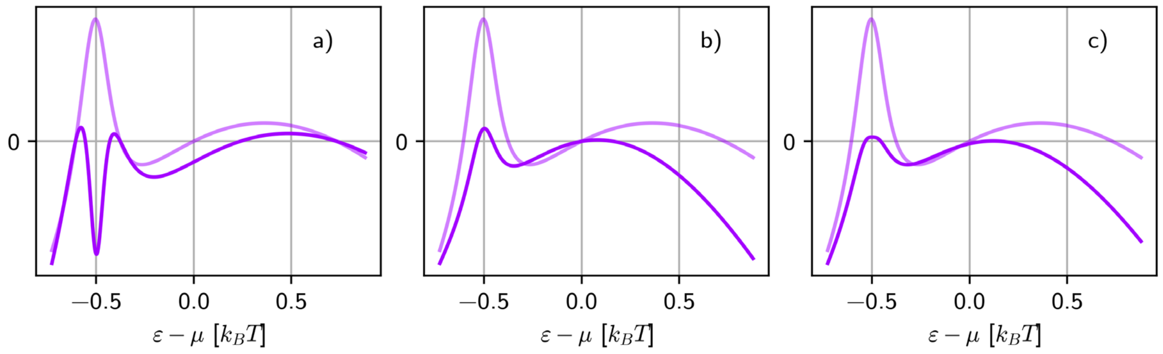



Effectively the two reservoirs are in equilibrium

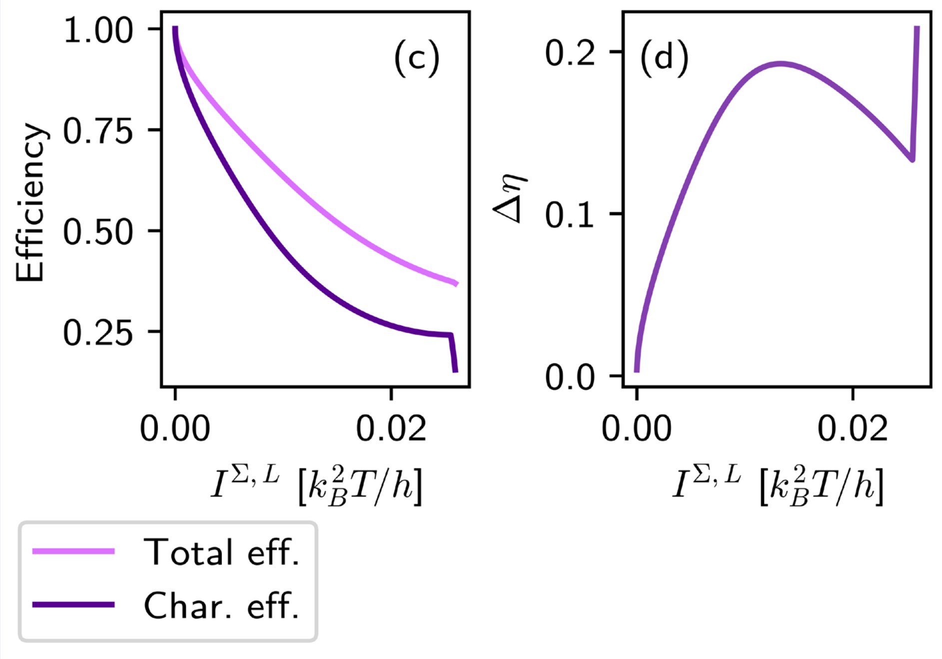
"Equivalent thermal distribution"

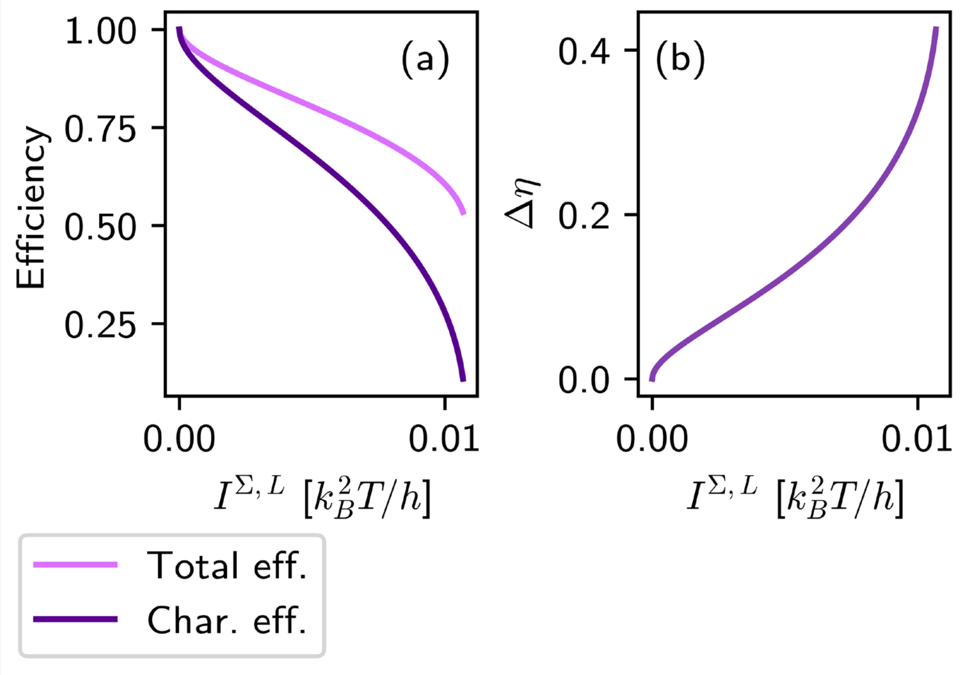
# Crossing points



# Relation between char. eff. and total



# Optimization with second variable